Module 10

Partha Pratim Das

Objectives & Outline

Set Operations

Null Values

Three Valued Logic

Aggregate Functions

Group By

Having

Null Values

Module Summary

Database Management Systems

Module 10: Introduction to SQL/3

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Module Recap

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Module Summary

- Completed the understanding of basic query structure

Module Objectives

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Module Summary

- To familiarize with set operations, null values and aggregation

Module Outline

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Module Summary

- Set Operations: union, intersect, except
- Null Values
- Aggregate Functions: avg, min, max, sum, and count
  - Group By
  - Having
  - Null Values

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Module Summary

- Find courses that ran in Fall 2009 or in Spring 2010

  (**select** *course_id* **from** *section* **where** *sem* = 'Fall' **and** *year* = 2009)
  union
  (**select** *course_id* **from** *section* **where** *sem* = 'Spring' **and** *year* = 2010)

- Find courses that ran in Fall 2009 and in Spring 2010

  (**select** *course_id* **from** *section* **where** *sem* = 'Fall' **and** *year* = 2009)
  intersect
  (**select** *course_id* **from** *section* **where** *sem* = 'Spring' **and** *year* = 2010)

- Find courses that ran in Fall 2009 but not in Spring 2010

  (**select** *course_id* **from** *section* **where** *sem* = 'Fall' **and** *year* = 2009)
  except
  (**select** *course_id* **from** *section* **where** *sem* = 'Spring' **and** *year* = 2010)

Set Operations (2)

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Module Summary

- Find the salaries of all instructors that are less than the largest salary

  select distinct $T.salary$
  from *instructor* **as** T, *instructor* **as** S
  where $T.salary < S.salary$

- Find all the salaries of all instructors

  select distinct *salary*
  from *instructor*

- Find the largest salary of all instructors

  (**select** "second query")
  except
  (**select** "first query")

Set Operations (3)

- Set operations **union**, **intersect**, and **except**
  - Each of the above operations automatically eliminates duplicates
- To retain all duplicates use the corresponding multiset versions **union all**, **intersect all**, and **except all**.
- Suppose a tuple occurs m times in r and n times in s, then, it occurs:
  - $m + n$ times in r **union all** s
  - $\min(m, n)$ times in r **intersect all** s
  - $\max(0, m - n)$ times in r **except all** s

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Module Summary

- It is possible for tuples to have a null value, denoted by *null*, for some of their attributes
- *null* signifies an unknown value or that a value does not exist
- The result of any arithmetic expression involving *null* is *null*
  - Example: $5 + null$ returns null
- The predicate **is null** can be used to check for null values
  - Example: Find all instructors whose salary is null

    select *name*
    from *instructor*
    where *salary* **is null**
- It is not possible to test for **null** values with comparison operators, such as $=$, $<$, or $<>$ We need to use the **is null** and **is not null** operators instead

Null Values (2): Three Valued Logic

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Module Summary

- Three values – **true, false, unknown**
- Any comparison with *null* returns *unknown*
  - Example: $5 < null$ or $null <> null$ or $null = null$
- Three-valued logic using the value unknown:
  - OR: (*unknown* **or** *true*) = *true*,
    (*unknown* **or** *false*) = *unknown*
    (*unknown* **or** *unknown*) = *unknown*
  - AND: (*true* **and** *unknown*) = *unknown*,
    (*false* **and** *unknown*) = *false*,
    (*unknown* **and** *unknown*) = *unknown*
  - NOT: (**not** *unknown*) = *unknown*
  - "*P* **is unknown**" evaluates to *true* if predicate *P* evaluates to *unknown*
- Result of **where** clause predicate is treated as *false* if it evaluates to *unknown*

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Module Summary

- These functions operate on the multiset of values of a column of a relation, and return a value

  avg: average value
  min: minimum value
  max: maximum value
  sum: sum of values
  count: number of values

Aggregate Functions (2)

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Module Summary

- Find the average salary of instructors in the Computer Science department

  select avg (*salary*)
  from *instructor*
  where *dept_name* = 'Comp. Sci';

- Find the total number of instructors who teach a course in the Spring 2010 semester

  select count (distinct *ID*)
  from *teaches*
  where *semester* = 'Spring' and *year* = 2010;

- Find the number of tuples in the *course* relation

  select count (*)
  from *courses*;

Aggregate Functions (3): Group By

- Find the average salary of instructors in each department

  select *dept_name*, **avg**(*salary*) **as** *avg_salary*
  from *instructor*
  group by *dept_name*;

ID	*name*	*dept_name*	*salary*
76766	Crick	Biology	72000
45565	Katz	Comp. Sci.	75000
10101	Srinivasan	Comp. Sci.	65000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000
12121	Wu	Finance	90000
76543	Singh	Finance	80000
32343	El Said	History	60000
58583	Califieri	History	62000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
22222	Einstein	Physics	95000

dept_name	*avg_salary*
Biology	72000
Comp. Sci.	77333
Elec. Eng.	80000
Finance	85000
History	61000
Music	40000
Physics	91000

Aggregate Functions (4): Group By

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Module Summary

- Attributes in **select** clause outside of aggregate functions must appear in **group by** list

  /* erroneous query */
  select *dept_name*, *ID*, **avg**(*salary*)
  from *instructor*
  group by *dept_name*;

Aggregate Functions (5): Having Clause

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Module Summary

- Find the names and average salaries of all departments whose average salary is greater than 42000

  select *dept_name*, *ID*, **avg**(*salary*)
  from *instructor*
  group by *dept_name*
  having avg(*salary*) $>$ 42000;

  Note: predicates in the **having** clause are applied after the formation of groups whereas predicates in the **where** clause are applied before forming groups

Null Values and Aggregates

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Module Summary

- Total all salaries

  select sum (*salary*)
  from *instructor*;

  - Above statement ignores null amounts
  - Result is *null* if there is no non-null amount
- All aggregate operations except **count(*)** ignore tuples with null values on the aggregated attributes
- What if collection has only null values?
  - count returns 0
  - all other aggregates return null

Module Summary

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Module Summary

- Completed the understanding of set operations, null values, and aggregation

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